Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_00a4e59a9f154dbfa61cfacf9e528d7c
Date	2026-11-02 15:30:54
Status	AI proposal, pending expert review

1. Problem

Defect	Severity	Confidence	Detail
size_mismatch	major	99%	length 114.52 mm (nom 120.00 ±0.10)
size_mismatch	major	99%	width 76.55 mm (nom 80.00 ±0.10)
size_mismatch	major	99%	hole dia 11.26 (nom 12.00 ±0.05), position off 1.66 mm (±0.10)
size_mismatch	major	99%	hole dia 11.41 (nom 12.00 ±0.05), position off 3.82 mm (±0.10)
size_mismatch	major	99%	hole dia 10.65 (nom 12.00 ±0.05), position off 2.61 mm (±0.10)
size_mismatch	major	99%	hole dia 10.78 (nom 12.00 ±0.05), position off 6.23 mm (±0.10)
dent	major	95%	~18 mm
dent	major	70%	~7 mm
dent	major	73%	~9 mm

2. Root cause

Cause: Tool Wear and Die Damage

Reason: The presence of multiple dents (18mm, 7mm, 9mm) and significant dimensional variations in hole diameters and positions strongly suggests degraded tooling surfaces and improper die alignment as described in [seed-003].

Effect: Deformation, Dimensional Inaccuracy

Confidence: 69%

Supporting evidence
- **seed-003**: Tool Wear and Die Damage cause scratches, dents and dimensional variations. Preventive actions: regular tooling inspection, preventive maintenance schedule, die regrinding.

3. Solution

Corrective actions (fix this occurrence)

1. Inspect die surfaces for damage
2. Regrind or replace worn tooling components

Preventive actions (stop it recurring)

1. Implement a regular tooling inspection schedule
2. Establish a preventive maintenance program for dies

Precautions

- Monitor for surface finish degradation during production runs

4. Other causes considered

Rank	Cause	Confidence	Reason
2	Improper Bending Force or Over Bending	66%	The major dimensional deviations in length and width (114.52mm vs 120mm; 76.55mm vs 80mm) indicate that the bending force or angle settings are incorrect, leading to significant dimensional inaccuracy [seed-001].
3	Material Inconsistency	63%	Variations in material hardness or properties can lead to unpredictable deformation and hole position shifts, which aligns with the observed out-of-tolerance hole positions [seed-002].

5. Traceability

AI model: gemini:gemini-3.1-flash-lite. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.